

KittyCheck: A Mobile Application for Automated Feline Grimace Scale Pain Detection

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Abstract—Pain assessment in cats is challenging due to them being good at hiding their discomfort, making early detection difficult. Although the Feline Grimace Scale (FGS) provides a standardized method for assessing pain in cats, many cat owners and veterinarians are still unfamiliar with its use. This study proposes *KittyCheck*, a region-specific convolutional neural network framework that automates FGS scoring. The system employs a multi-stage pipeline: EfficientNetB0-based face detection, region detection, and five independent CNN classifiers trained on a dataset of 1,598 cat images. Each facial action unit scorer produces scores from 0–2, which are aggregated into a total FGS score and binary pain classification (threshold ≥ 4). The models are deployed within the Android application for on-device inference. Experimental results show a full pipeline pain classification accuracy of 87.0% (F1=84.2%), a mean absolute error of 0.83 FGS points, and 91.3% of predictions within 2 points of ground truth. Usability evaluation with 20 participants yields an overall System Usability Scale score of 81.75, indicating good accessibility for normal cat owners and veterinarians. This research demonstrates a practical and interpretable mobile system for automated feline pain assessment.

Index Terms—Feline Grimace Scale, pain detection, convolutional neural networks, region-specific classification, computer vision, mobile application, animal welfare

I. INTRODUCTION

A. Background of the Study

The worldwide increase in cat ownership means that there will be more cats in need of veterinary care. Majority of them will undergo at least one surgery and many will have other medical conditions that can result in acute pain [1]. Unlike dogs, cats display only subtle behavioral or facial cues when in pain, which is harder to detect even for trained professionals. This results to pain in cats undiagnosed until symptoms become more obvious, prolonging their suffering that could lead to worsening their medical condition [2]. This highlights the need for an objective, reliable, and accessible method for assessing pain in cats.

While there are multiple pain assessment tools created for cats, many of them are designed for a specific use-case or needs clinical observations, which limits their general use. An example for this are: UNESP-Botucatu Multidimensional Composite Pain Scale for Assessing Postoperative Pain in Cats [3] and Glasgow Feline Composite Measure Pain Scale [4] for clinical evaluation. So to provide a standardized alternative, Steagall and his team developed the Feline Grimace Scale

(FGS) in 2019. The FGS evaluates five facial action units (FAUs) which are: ear position, orbital tightening(eyes), muzzle tension, whiskers change, and head position to assess acute pain [5]. This tool has been validated by multiple studies for its reliability in acute pain assessment [6]. However, this tool still requires manual scoring, time, and some training, making veterinarians hesitant to use it despite knowing of its existence [7].

Automating task such as pain assessment using facial expressions has already been proven to be successful in humans through machine learning [8]. And researchers have adopted similar approaches for feline pain assessment. In a 2023 study, Feighelstein et al. [9] compared two approaches in automating pain recognition in cats; their result showed that the landmark-based approach achieved an accuracy above 77%, while the deep learning approach reached an accuracy above 65%. Landmark-based approached explicitly detect and mark predetermined facial points, which could help interpret how the results are derived. However, this approach is sensitive to the environmental conditions of the data, being affected by lighting, pose, or fur patterns. In contrast, CNN-based models learn relevant features of cats directly from image pixel, it is good for capturing subtle and complex patterns that may be overlooked by landmarks. However, they are less interpretable and requires significantly larger dataset to achieve high accuracy, while landmark-based approaches perform well in smaller, but accurately annotated datasets. Both approaches showed strong potential for automated pain detection [9].

In a similar study, Steagal et. al [10] developed hybrid approach combining deep learning models for facial landmark prediction and ensemble learning models for predicting FGS scores. Their system achieved high accuracy (up to 95.5%). And one of their relevant findings is that models using ShuffleNetV2, EfficientNetB0, and MobileNetV3 architectures performed the best in predicting landmarks, these AI models are known to be efficient and have low computational requirement, highlighting the potential for a lightweight mobile application.

B. Statement of the Problem

Cats are very good at concealing their pain, making early detection challenging for cat owners and even veterinarians. Existing pain assessment tools such as the Feline Grimace Scale are highly reliable but still require some training, time, and manual scoring. Although machine learning approaches show promise in automating this process, they are still in their early stages and not accessible to the public. This just highlights the need for a practical and user-friendly mobile application that automatically assesses pain in cats, providing

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a reliable tool for cat owners to monitor the well being of their cats and support timely veterinary care.

C. Objectives

The main objective of the study is to develop a user friendly mobile application that can automatically detect pain in cats in real time based on the Feline Grimace Scale (FGS). More specifically, this research aims:

- To develop a deep learning system that integrates face detection, facial region localization, and five independent facial action unit classifiers to generate FGS scores from cat images.
- To evaluate the usability and user experience of the mobile application among cat owners, veterinary students, and veterinarians.
- To assess the accuracy and reliability of the model by comparing its predictions with expert-annotated FGS scores.

D. Significance of the Study

This study will contribute to the fields of animal welfare, and veterinary science, and applied artificial intelligence by providing a proof-of-concept system for automated and objective pain detection in cats. The potential beneficiaries will include:

- Pet owners and cat caregivers, who will use the system as an accessible tool to better understand their cats' condition, monitor their well-being, and seek timely veterinary care;
- Veterinarians, who may use the system to complement clinical judgment and improve diagnostic accuracy;
- Researchers, who can build upon this region-focused deep learning framework to explore further applications of artificial intelligence in animal health and welfare.

E. Scope and Limitations

This study will focus on cats and will use the Feline Grimace Scale (FGS) to assess pain from static images of cat faces, detecting and scoring the predefined facial action units associated with pain. Another limitation of the study is that the datasets used for training were primarily obtained from publicly available image sources rather than controlled clinical environments. The project will implement CNN-based models trained on annotated images, and pain detection will be limited to the categories defined by the FGS. The study will not include real-time video monitoring or continuous tracking, and will not assess other species or pain scales beyond FGS. Additionally, the application is designed for and evaluated within the Android environment.

II. REVIEW OF RELATED LITERATURE

A. The Feline Grimace Scale (FGS)

In 2019, Evangelista et al. [5] developed and validated the Feline Grimace Scale (FGS) in response to the limited availability of a grimace-based pain assessment scoring tools

for cats. The tool evaluates acute pain using five facial action units (FAU): ear position, orbital tightening, muzzle tension, whisker position, and head position. Although some FAUs were easier to score than others, with whiskers being the hardest ($ICC_{\text{single}} = 0.55\text{--}0.60$), while eyes ($ICC_{\text{single}} = 0.83\text{--}0.86$) and ears being easiest ($ICC_{\text{single}} = 0.85\text{--}0.87$). Overall, the scale showed good inter-rater reliability ($ICC = 0.89$) and excellent intra-rater reliability ($ICC > 0.91$). The researchers found that painful cats scored higher on the FGS than pain-free cats, and that scores decreased after pain treatments showing that the tool responds to changes in pain level. The FGS also scored closely to an established veterinary pain scale, Glasgow Composite Measure Pain Scale–Feline (CMPS-Feline) [4]. The tool was later shown to be valid to use even in kittens [11]. Overall, this shows that Feline Grimace Scale is a valid and reliable tool for assessing acute pain in cats.

To further study the validity of the Feline Grimace Scale, Evangelista and Steagall [12] examined the agreement and reliability of the tool among groups with different levels of expertise: cat owners, undergraduate veterinary students, nurses, and veterinarians. Their results showed that all groups had good inter-rater reliability for the final FGS scores, while students and veterinarians had excellent intra-rater reliability, and owners and nurses had good reliability. This provides evidence that even non-veterinarians can reliably use the Feline Grimace Scale. To further support this, a global survey conducted by [13] with 1,540 cat caregivers from 66 countries showed that no demographic factors significantly affected FGS scoring. Cat caregivers' total FGS scores were very similar to veterinarians' scores, except slightly higher in muzzle tension. This indicates that cat caregivers have good reliability in using FGS regardless of their level of expertise, age, gender, or country.

In both studies, the participants received minimal training in the form of the official Feline Grimace Scale training manual (<http://www.bit.ly/FGSmanual>). The effects of training on FGS scoring were further studied by Robinson and Steagal [2], who found that inter-rater reliability and agreement between veterinarians and expert raters improved after the training. Based on this, the researchers advised veterinarians to provide basic training in FGS scoring when using the tool in clinical practice to help standardize feline pain assessment. The importance of training could also be a major factor in the positive results found in both Evangelista and Steagall's [12] and Monteiro et al.'s [13] studies.

These studies demonstrate that the Feline Grimace Scale is a valid and reliable tool for assessing pain in cats. It can be used by veterinarians in clinical settings and even normal cat owners to detect acute pain early. However, even with all of the existing studies, the use of validated tools in veterinary practice remains low. In a survey among Spanish small animal veterinarians, despite 82% of 292 participants reporting the awareness to pain assessment scales, only 28% of them actually use the tools. And among the veterinarians (72%) not using scales, the main limiting factors are the lack of training, limited time, and insufficient personnel [7]. Thus, there is a need to make FGS more accessible, support to its training, and reduce the time required for manual scoring.

B. Methods for Automated Pain Detection

In 2022, Feighelstein et al. [14] were the first to apply study automation in pain recognition in cats, using 29 domestic shorthair cats (464 images) undergoing ovariohysterectomy (spay) as their dataset. In their study, they compared two approaches: landmark-based approach (LDM) using 48 facial landmarks related to Cat Facial Action Coding System (CatFACS) [15]; and deep learning (DL) model based on ResNet50 CNN fine-tuned for binary classification (pain/no pain). Both their LDM and DL models achieved above 72% accuracy in detecting pain from cat faces, which indicates the potential of both approaches as basis for automated pain recognition in cats. However, as the study was still on proof-of-concept stage, several limitations remained, such as the lack of larger and more diverse cat populations, the labor-intensive manual annotation of 48 landmarks, and the CNN model lacking interpretability.

As an extension of this work, Feighelstein et al. [9], focused on incorporating explainable AI on the two approaches to identify which facial regions are the most important in pain classification. They also used a more heterogeneous dataset with 84 client-owned cats of different breeds, ages, sex, and their pain indication were scored by veterinarians using CMPS-Feline [4]. Their results showed that LDM approaches generalize better with smaller noisier datasets achieving 77% accuracy compared to 65% accuracy for the ResNet50 model. The decrease in accuracy for DL was most likely due to the much smaller dataset of the current study compared to the 2022 study. By using explainable AI, they found that the mouth and nose regions were the most important for the machine in recognizing pain, while ears were the least important. These findings suggest that the machines rely on specific facial regions rather than the entire face, supporting the idea that region-focused models may be more efficient or accurate than full-face or landmark pipelines.

Landmark-based approaches are the most common method in feline pain detection, as CatFACS [15] had already been established even before the first attempts to automate this process. To further advance the research that use this framework, Martvel et al. [16] introduced the Cat Facial Landmarks in the Wild (CatFLW) dataset and develop a landmark detection CNN-based model which can predict the 48 CatFACS-related landmarks. Their system uses a three-stage architecture using EfficientNetV2 as the backbone consisting of the initial face detection, followed by a region detection, and then the ensemble detection that outputs the coordinates of all 48 landmarks. Martvel et al. [17] later used this landmark detection system to train a pain recognition model and compare its performance against models that uses manually annotated landmarks. Utilizing the same cats-in-pain dataset from Feighelstein et al. [14], results showed that the accuracy reached only 66% for the automatically detected landmark, while the manual annotations almost have the same accuracy from the previous study reaching 73%. Although automating the detection of cat facial landmarks is a huge advancement in automating the pain detection of cats, it comes with a reduction in accuracy, highlighting that landmark-based pipelines are more

vulnerable to localization errors compared to classic deep learning models.

In a recent study, Pertpraiwong and Kangkachit [18] used a similar approach in assessing pain level in cats. However, instead of the established 48 landmarks related to CatFACS, the researchers used YOLOv8-Pose to detect 37 facial keypoints with the goal of evaluating the pain levels in cats based on the Feline Grimace Scale. The keypoint detection model achieved high accuracy ($mAP50=0.995$) and pain classification accuracy reaching 87%. The highest keypoint accuracy was observed in the eyes (NRMSE= 2.24%) and muzzles (NRMSE=2.72%), while the lowest was observed in whiskers (NRMSE=22.53%) and shoulder regions (NRMSE=21.21–20.60%), reflecting the inherent difficulty of detecting small features and variable visibility.

Having smartphone applicability in mind, Steagall et al. [10] developed a fully automated deep learning system for prediction of pain using Feline Grimace Scale. Similar to Pertpraiwong and Kangkachit (2025), they used CNN models for predicting 37 landmarks and found that the best-performing CNN backbones were ShuffleNetV2, EfficientNetB0, and MobileNetV3, with NRMSE values around 16–18% when combined with face alignment and Laplacian edge detection. Using geometric descriptors from the predicted landmarks, XGBoost models then classifies painful vs. non-painful states of cats, reaching accuracy up to 95.5%. The prediction speed and model sizes of their system is suitable for smartphone apps. However, this approach comes with various limitations, including the need for semi-automatic pre-annotation of 10 of the 37 landmarks for face alignment, the absence of cat face detector, and large prediction errors in whisker change and head position landmarks due to inconsistencies in manual annotation.

Across existing literature, the current automated cat pain detection methods are dominated by landmark-based pipelines, with limited exploration of pure deep learning-based approaches. Although newer studies have shown that landmark detection can produce accurate pain classifications, these methods have difficulty capturing subtle facial regions and need manual or semi-automatic preprocessing steps. In contrast, the Feline Grimace Scale is a system that evaluates specific facial action units rather than the whole face of cats, yet no study has explored a region-specific CNN framework that directly scores each AU without relying on full landmark prediction. This represents a clear research gap. A CNN-driven parts-based approach could offer interpretable results to each facial action units of FGS, reduce the dependence on dense landmark detection, and with lightweight architecture, feasible for real-time mobile deployment.

III. METHODOLOGY

A. Development Environment

1. System Requirements

The deep learning models and mobile application were developed using a system with the following specifications:

- Operating System: Windows 11 Pro 64-bit
- Processor: AMD Ryzen 7 6800HS

- Memory: 16 GB DDR5 RAM
- GPU: NVIDIA GeForce RTX 3060
- Storage: 1 TB NVMe SSD

The mobile application was tested using an Android emulator with the following configuration:

- Device: Google Pixel 6 (Emulator)
- Android Version: Android 14 (API 34)

2. Development Tools and Technologies

Various tools and technologies were utilized in the development of the deep learning models and mobile application. The following were used for model training:

- PyTorch: A deep learning framework used for training the models
- Torchvision: A library used for image transformations and dataset handling
- TIMM: A library providing pre-trained EfficientNet models
- OpenCV: A library used for image processing
- Scikit-learn: A library used for evaluation metrics and data splitting
- Matplotlib: A library used for visualizing training results

The following tools and technologies were used for mobile application development:

- Flutter: A framework used for developing the mobile application interface
- Camera: A plugin used for capturing images within the application
- Image Picker: A plugin used for selecting images from the device gallery
- SQLite: A plugin used for local database storage

The following tools were used for model conversion and deployment:

- Torch ONNX: A tool used for exporting PyTorch models to ONNX format
- ONNX: An intermediate representation format for model conversion
- ONNX2TF: A tool used for converting ONNX models to TensorFlow Lite format

3. Development Environment

The project was developed using the following integrated development environments:

- Visual Studio Code: The primary code editor used for Python and Flutter development
- Android Studio: Used for emulator management and mobile application testing

B. Research Design

This research employed an experimental and developmental research design aimed at designing, implementing, and evaluating a mobile-based automated system for feline pain detection using the Feline Grimace Scale (FGS). The research followed a quantitative approach, utilizing deep learning techniques to analyze facial features of cats and classify pain levels based on established FGS facial action units. The development process adhered to an iterative pipeline architecture consisting of three stages: (1) facial detection and localization, (2)

facial action unit region identification, and (3) individual FGS scoring followed by aggregation. Each stage was developed, trained, and validated independently before integration into a unified mobile application for real-world deployment.

C. System Architecture Overview

The proposed system, *KittyCheck*, follows a modular architecture composed of a deep learning backend for automated Feline Grimace Scale (FGS) scoring and a Flutter-based mobile application for user interaction and on-device inference. All image processing is performed locally to ensure privacy and offline usability.

The workflow begins with image capture or upload via the mobile application. The input image undergoes face detection, face region detection, alignment, and region extraction. Each extracted region is then processed by lightweight CNN models to generate facial action unit (FAU) scores.

The final Feline Grimace Scale (FGS) score is computed by aggregating the individual facial action unit (AU) scores. Pain classification is then determined using a threshold, where scores below 4 indicate no pain, while scores of 4 and above indicate pain. The results are displayed to the user, including AU predictions and an option for manual score adjustment.

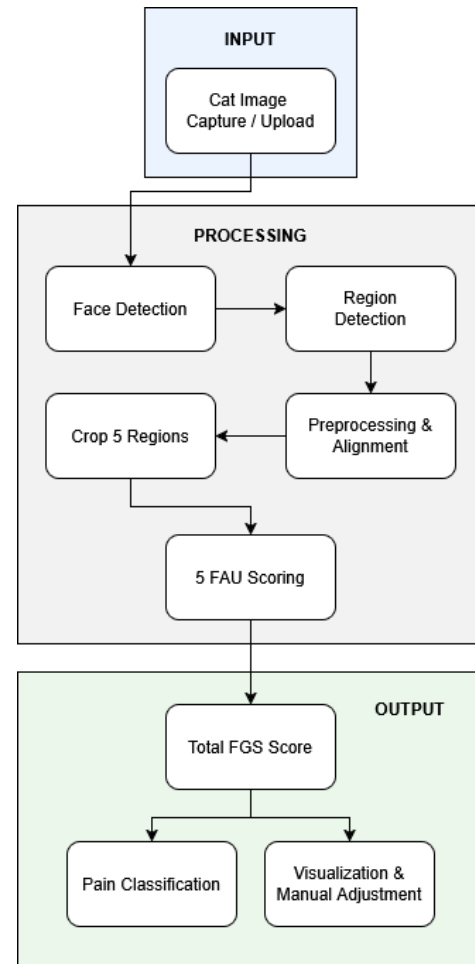


Fig. 1: System Architecture

D. Dataset Preparation

1) *CatFLW Dataset (Detection Models)*: The *Cat Facial Landmarks in the Wild (CatFLW)* dataset [19] was used to train the face and region detection models. It contains approximately 2,079 images of cats in diverse conditions, annotated with facial bounding boxes and 48 landmark points (eyes, ears, nose, muzzle, and whisker pads). These annotations provided precise ground-truth coordinates for extracting facial regions required for Feline Grimace Scale (FGS) scoring.

2) *Manual FGS Annotation*: A separate dataset of 1,598 images was manually annotated with Feline Grimace Scale (FGS) scores using images selected from the CatFLW dataset [19], Crawford Cat Dataset [20], and other public sources.

Annotations were performed in reference to the official Feline Grimace Scale (FGS) manual. Each image was independently annotated by the proponent, with assistance from two additional annotators. The final facial action unit (FAU) scores was determined using the median of the three annotations to ensure consistency. Annotations were then stored in a CSV file containing image paths and FAU scores.

The dataset distribution and train/validation/test splits are summarized in Table I. The dataset was divided using a 70%–15%–15% split for training, validation, and testing, respectively, ensuring that each Action Unit maintained consistent class proportions across all subsets.

TABLE I: Dataset distribution and train/validation/test splits per Action Unit (AU)

AU	Total	Class 0	Class 1	Class 2	Train	Val/Test
Ears	1593	853	538	202	1115	239 / 239
Eyes	1598	1072	358	168	1118	240 / 240
Muzzle	1541	708	616	217	1077	232 / 232
Whiskers	1520	732	594	194	1063	229 / 228
Head	1461	804	439	218	1021	220 / 220

All splits were stratified by FAU score to preserve class distribution, using a fixed random seed (42) for reproducibility.

E. Data Preprocessing Pipeline

The preprocessing pipeline prepared raw cat images for automated Feline Grimace Scale (FGS) scoring. First, the face detector identified and cropped the cat's face from the input image. The cropped face was then processed by the region detector, which identified five key facial points corresponding to the eyes, nose, and ears. These points were used to align the face and guide the extraction of facial regions.

To make the images more consistent, the face was rotated so that the eyes were horizontally aligned. After alignment, five facial regions were cropped from the image: ears, eyes, muzzle, whiskers, and head position. Each cropped region was resized to 224×224 pixels before being passed to the lightweight CNN models for FGS scoring.

F. Model Development – Detection Models (Face & Region)

The face and region detection models were based on Martvel et al. [19], using the CatFLW dataset for training. Both models adopt an EfficientNetB0 backbone with custom regression heads for coordinate prediction.

The face detector predicts the bounding box of the cat's face from a 224×224 input image. It consists of a pretrained EfficientNetB0 backbone with the classification head removed, followed by three fully connected layers with 128, 64, and 4 units. The model was trained using mean squared error (MSE) loss and the Adam optimizer with a learning rate of 1×10^{-4} . Data augmentation techniques such as flipping, rotation, color jitter, blur, and noise were applied to improve generalization.

The region detector predicts five facial landmarks (eyes, nose, and ears) derived from the CatFLW annotations. It uses the same architecture as the face detector, except for an output layer of 10 units corresponding to five (x, y) coordinate pairs. It was trained using MSE loss and the Adam optimizer, and uses face crops generated by the face detector instead of ground-truth boxes to better reflect real-world conditions.

Both models were trained on the CatFLW dataset for accurate face localization and landmark estimation.

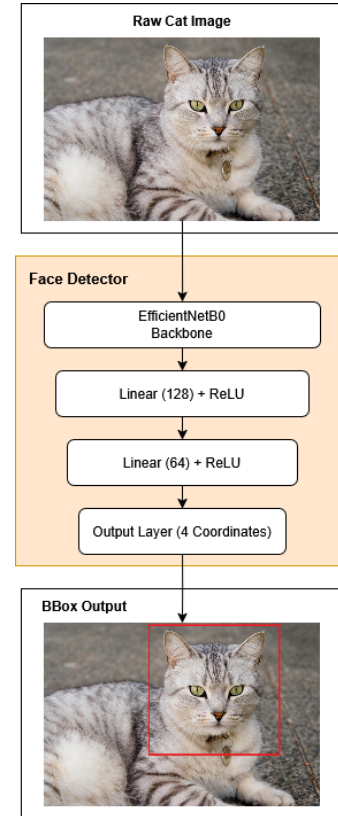


Fig. 2: Architecture of the EfficientNetB0-based face detector used to localize and crop the cat face region.

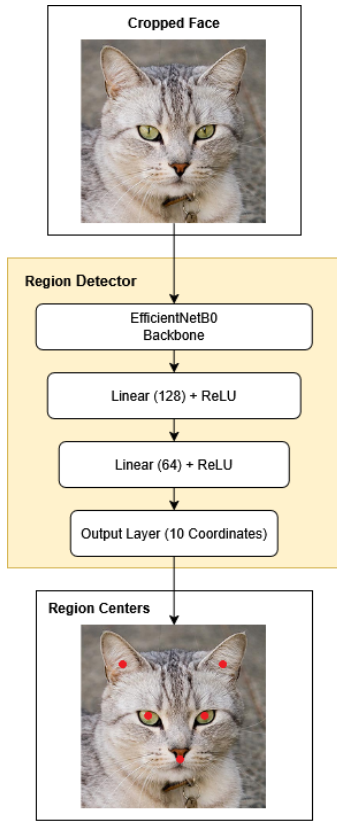


Fig. 3: Architecture of the EfficientNetB0-based region detector used to predict five facial region centers from the cropped face image.

G. Model Development – FGS Scorers

Five independent EfficientNetB0-based classifiers were developed to score each Feline Grimace Scale (FGS) Action Unit (AU): ears, eyes, muzzle, whiskers, and head position. As shown in Figure 4, all models shared the same backbone architecture and classification head design, but were trained independently using their corresponding cropped facial regions. Each model received a $224 \times 224 \times 3$ input image and predicted one of three pain scores (0, 1, or 2).

All classifiers were trained using the stratified 70%–15%–15% train/validation/test splits described in the dataset preparation stage. To address class imbalance, class weights were applied during training and minority classes were oversampled using a `WeightedRandomSampler`. The models were trained using the Adam optimizer with `ReduceLROnPlateau` scheduling, gradient clipping, and early stopping to improve training stability and reduce overfitting.

Data augmentation techniques were also applied to improve model generalization. These included horizontal flipping, rotation, color jittering, Gaussian blur, additive noise, and random perspective transformations. Augmentation intensity and regularization settings were adjusted per AU to account for differences in feature complexity and class imbalance severity across facial regions.

Additional hyperparameter tuning was performed for each FAU scorer to account for differences in feature complexity and class imbalance across facial regions. The ears and eyes

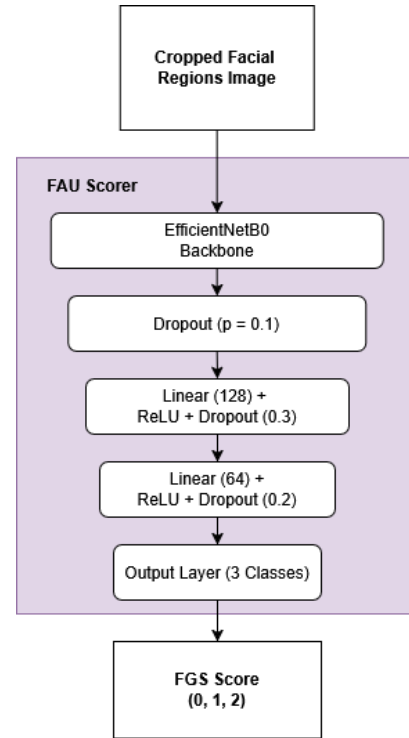


Fig. 4: Shared EfficientNetB0-based architecture used for the independent FGS Action Unit scorers.

models used moderate regularization settings, while the muzzle and whiskers models applied stronger augmentation due to their more subtle facial features and higher class imbalance. The head position scorer additionally used label smoothing because it relied on broader facial context instead of localized facial regions.

H. Mobile Deployment

The trained PyTorch models were converted into TensorFlow Lite format to enable efficient on-device inference. This conversion allows the models to run directly on mobile devices without requiring cloud computation. All seven models, consisting of one face detector, one region detector, and five Action Unit (AU) scorers, were integrated into the Flutter application as local assets.

1) *Mobile Application Overview:* The *KittyCheck* mobile application was developed to provide a full pipeline for feline pain assessment, from image acquisition to result interpretation and record management.

Application Workflow Screens:

- **Onboarding Screen:** Introduces the Feline Grimace Scale and provides usage instructions.
- **Home Screen:** Serves as the main entry point for image capture or gallery selection.
- **Camera/Gallery Screen:** Allows users to capture images using the device camera or select from local storage.
- **Results Screen:** Displays predicted Action Unit scores with corresponding visual indicators and computed FGS score.

- **Manual Editing Screen:** Enables user adjustment of automatically predicted AU scores.
- **History Screen:** Stores and manages previous assessments with delete, search, and sorting functionality.

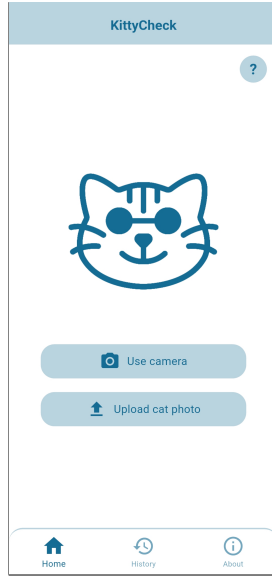
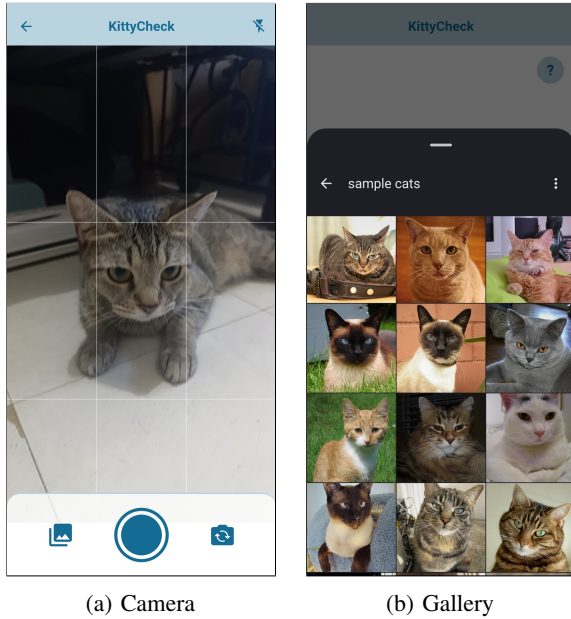


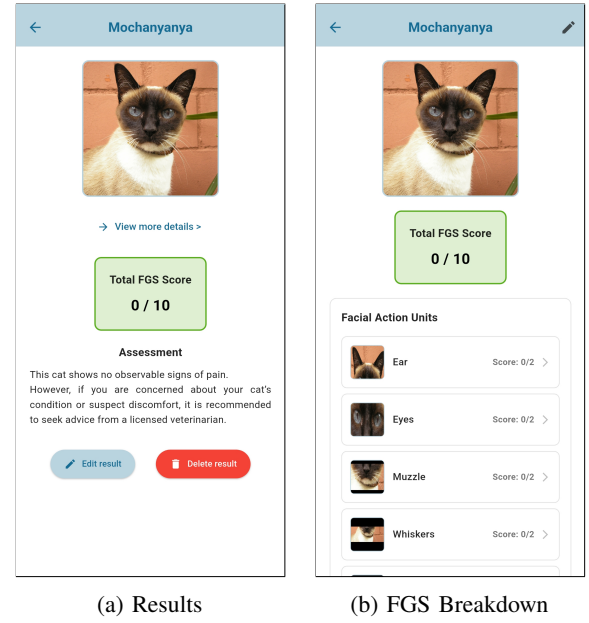
Fig. 5: Home Screen



(a) Camera

(b) Gallery

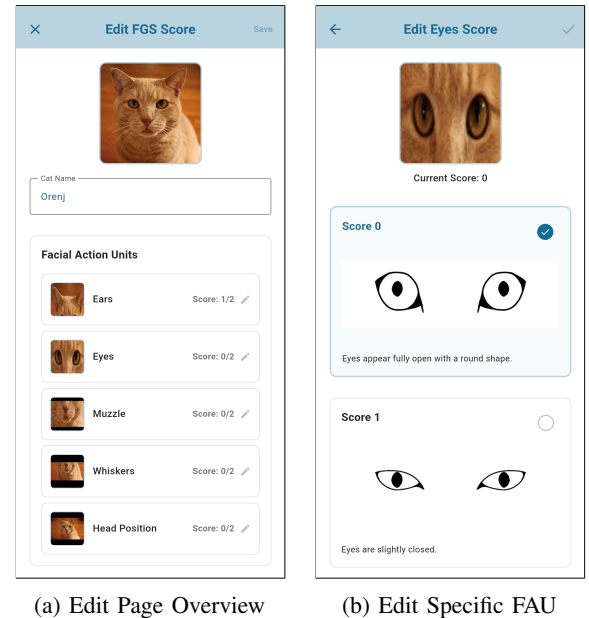
Fig. 6: Image acquisition interfaces.



(a) Results

(b) FGS Breakdown

Fig. 7: FGS prediction results and score breakdown.



(a) Edit Page Overview

(b) Edit Specific FAU

Fig. 8: Manual score adjustment interfaces.

I. Model Evaluation and Performance Metrics

1) *Detection Models (Face and Region)*: The face detection model was evaluated using Mean Squared Error (MSE) between predicted and ground truth normalized bounding box coordinates, as well as Intersection over Union (IoU) to measure localization accuracy on entire CatFLW dataset.

The region detection model was evaluated using MSE on normalized (x, y) coordinates of the five facial landmarks.

2) *FGS Classification Models*: Each FAU classifier was evaluated on a held-out test set to assess real-world performance. Performance metrics included accuracy, precision, recall, and weighted F1-score to account for class imbalance. Confusion matrices were used to analyze misclassification

patterns across classes.

3) *Full Pipeline System Performance*: System-level performance was evaluated using the final aggregated Feline Grimace Scale (FGS) output.

FGS Score Error: Mean Absolute Error (MAE) between predicted and ground truth total FGS scores, along with within-1-point and within-2-point accuracy.

Pain Classification Performance: Binary classification performance was measured using accuracy, precision, recall, and F1-score. The system classifies pain as: scores below 4 indicate no pain, while scores of 4 and above indicate pain.

J. System Usability Evaluation

The usability of the *KittyCheck* mobile application was evaluated using the System Usability Scale (SUS). This standardized questionnaire was used to assess user experience in terms of ease of use, learnability, and overall system satisfaction.

The participants consisted of cat owners, veterinary students, and veterinarians. Each participant was asked to use the application before completing the SUS questionnaire.

The SUS responses were collected using Google Forms. All items were presented using a 5-point Likert scale, and the final SUS score was computed following the standard scoring procedure.

IV. RESULTS AND DISCUSSION

This section presents the performance of the proposed *KittyCheck* system, including detection accuracy, Feline Grimace Scale (FGS) scoring performance, full pipeline pain classification, and usability evaluation.

A. Detection Model Performance

The proposed pipeline consists of two detection stages: face localization and facial region detection. The face detection model achieved a mean squared error (MSE) of 0.000437 on normalized bounding box coordinates and an Intersection over Union (IoU) of 0.8526 on the entire CatFLW dataset, indicating accurate localization of feline faces across varying image conditions.

The region detection model was evaluated using both ground-truth and predicted face bounding boxes to assess robustness in real-world deployment. Using ground-truth face crops, the model achieved a mean pixel error of 5.72 pixels across the five facial regions. When evaluated using predicted face bounding boxes from the face detector, the mean pixel error remained comparable at 5.69 pixels, suggesting minimal error propagation between pipeline stages.

TABLE II: Detection Model Performance

Model	MSE	IoU	Mean Pixel Error
Face Detector	0.000437	0.8526	-
Region Detector (GT bbox)	0.000750	-	5.72 px
Region Detector (Pred bbox)	0.000953	-	5.69 px

These results indicate that the detection pipeline can reliably localize facial regions required for FGS assessment. The

small difference between ground-truth and predicted bounding box performance demonstrates that the face detector provides sufficiently accurate crops for downstream region scoring. The strong performance is consistent with the detection framework adopted in this study, originally proposed by Martvel et al. [19].

B. FGS Scorer Performance

Five independent EfficientNetB0-based classifiers were trained to predict scores for each Feline Grimace Scale Action Unit (AU): ears, eyes, muzzle, whiskers, and head position. Each model was evaluated on a held-out test set to assess real-world performance.

TABLE III: FGS Scorer Test Set Performance

AU	Accuracy	F1-Score	Precision	Recall	Support
Ears	78.2%	0.78	0.83	0.85	239
Eyes	85.8%	0.86	0.94	0.90	240
Muzzle	66.4%	0.65	0.66	0.66	232
Whiskers	69.3%	0.69	0.71	0.69	228
Head	68.2%	0.69	0.71	0.68	220

Among all facial regions, the eyes and ears achieved the highest performance, with accuracies of 85.8% and 78.2%, respectively. These regions are visually prominent and generally exhibit clearer expression patterns associated with pain, which may explain their stronger performance.

In contrast, muzzle, whiskers, and head position showed notably lower accuracy. The lower performance on these subtle facial regions aligns with the known difficulty of detecting fine-grained expressions and the subjective nature of scoring certain FAUs.

The results suggest that the proposed region-specific approach can effectively learn discriminative pain-related facial features without relying on dense landmark prediction, though performance varies across facial regions. The modular architecture allows each facial action unit scorer to operate independently, improving interpretability and enabling targeted retraining for underperforming regions such as whiskers and muzzle.

C. Full Pipeline System Performance

The complete *KittyCheck* pipeline was evaluated on an expert-annotated test set of 23 images to assess pain classification and FGS score prediction performance in a real-world evaluation scenario.

TABLE IV: Full Pipeline Pain Classification and FGS Scoring Performance on Expert-Annotated Test Set

Metric	Expert-Annotated Test Set
Accuracy	87.0%
Precision	100.0%
Recall	72.7%
F1-Score	84.2%
MAE (FGS)	0.83 points
Within 1 pt	78.3%
Within 2 pts	91.3%

The proposed system achieved an overall pain classification accuracy of 87.0% on the expert-annotated test set, with

a recall of 72.7%. The high precision (100.0%) indicates that when the system predicts pain, it is highly confident in its assessment, while the recall suggests there is room for improvement in capturing all painful cases.

TABLE V: Confusion Matrix for Pain Classification on the Expert-Annotated Test Set

	Predicted No Pain	Predicted Pain
Actual No Pain	12 (TN)	0 (FP)
Actual Pain	3 (FN)	8 (TP)

For FGS score prediction, the system achieved a mean absolute error (MAE) of 0.83 points on the expert-annotated test set. The system predicted scores within 1 point of the ground truth in 78.3% of cases and within 2 points in 91.3% of cases.

D. Per-Region Full Pipeline Accuracy

To further evaluate facial action unit performance within the complete pipeline, per-region accuracy was measured during a full pipeline testing on the expert-annotated test set.

TABLE VI: Facial Action Unit Accuracy on Expert-Annotated Test Set

Region	Correct	Total	Accuracy
Ears	19	23	82.6%
Eyes	21	23	91.3%
Muzzle	16	23	69.6%
Whiskers	15	23	65.2%
Head	17	23	73.9%

The full pipeline evaluation produced trends consistent with the individual classifier performance. Eyes and ears remained the most reliable indicators, while muzzle, whiskers, and head position showed notably lower accuracy. The performance variation across facial regions highlights the differing difficulty levels of scoring each facial action unit, with whisker and muzzle regions requiring further improvement. The modular multi-region scoring strategy remains robust even when individual regions underperform, as pain assessment aggregates multiple facial action units.

E. System Usability Evaluation

The usability of the *KittyCheck* mobile application was evaluated using the System Usability Scale (SUS). A total of 20 participants completed the evaluation, including cat owners, veterinarians, and veterinary students.

TABLE VII: SUS Survey Results

User Group	N	Mean SUS Score
Cat Owners	16	82.5
Veterinarians	2	73.75
Veterinary Students	2	83.75
Overall	20	81.75

The application achieved an overall SUS score of 81.75, which is considered above average and indicates good usability. Participants generally reported that the application was easy to use and effective for automated feline pain assessment.

Survey responses from 20 participants, including cat owners, veterinarians, and veterinary students, further indicated that users found the application intuitive and accessible for automated feline pain assessment. Participants particularly appreciated the clean interface, reference illustrations, and straightforward scoring presentation. However, several respondents reported challenges related to image capture conditions such as lighting and camera angle, and occasional lag during analysis. Some users also suggested improved error handling for failed image captures. The feedback therefore highlights several directions for future improvement, including enhanced image quality validation, interactive user guidance during image capture, improved processing responsiveness, and additional training data for difficult facial regions.

Overall, the experimental results demonstrate that the proposed region-specific CNN pipeline can effectively automate Feline Grimace Scale assessment while maintaining strong classification performance and interpretability. The combination of reliable facial region localization, independent FAU scoring, and high usability supports the feasibility of the proposed system as a practical tool for feline pain assessment.

V. CONCLUSION AND FUTURE WORK

This study demonstrated the feasibility of automated feline pain assessment using a region-specific convolutional neural network approach based on the Feline Grimace Scale. The proposed *KittyCheck* system achieved a full pipeline pain classification accuracy of 87.0% with a mean absolute error of 0.83 FGS points on expert-annotated test data, indicating promising performance for automated pain assessment. The results further showed that region-specific scoring can effectively capture pain-related facial expressions while maintaining interpretability through independent facial action unit assessment. In addition, the mobile application achieved a high usability score, suggesting that the system is accessible to both veterinary professionals and general cat owners. Overall, the proposed system provides a practical and interpretable framework for automated feline pain assessment and demonstrates potential as a supportive tool for clinical decision-making and improving animal welfare.

Despite encouraging results, the system should currently be viewed as a supportive assessment tool rather than a replacement for professional veterinary evaluation. Whisker and muzzle regions demonstrated lower performance compared to other facial action units, suggesting that additional training data and improved localization strategies may be necessary for fine-grained facial features. Further validation across different breeds, lighting conditions, camera angles, and clinical environments is also necessary to improve system generalizability. Future work should focus on enhancing performance for underperforming regions, improving image capture guidance, adding automatic image quality validation, and optimizing for faster on-device mobile inference.

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